

precog-esgf-intake: Automated discovery, validation, and optimized download management of Earth system model data from Earth System Grid Federation nodes.

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Summary

precog-esgf-intake is a Python command-line-interface (CLI) software for automated discovery, checking, validation, and download of Earth System Model outputs from Earth System Grid Federation (ESGF) archives. The software is designed for research workflows that require reproducible access to large, distributed climate-model datasets and is particularly aimed at bulk screening of Earth System archives before downstream analysis. The software builds on inherited ESGF access functionality from intake-esgf (Collier et al., 2026), while introducing interactive CLI workflows for archive interrogation, file-availability checks, grid-consistency cross-checks, temporal validation, export of search results in tabular formats, and optimised download management of shortlisted Earth System Model data products.

Statement of need

Modern Earth System science workflows frequently rely on climate-model data distributed across Earth System Grid Federation (ESGF) nodes (Cimadevilla et al., 2025). For example, a Coupled Model Intercomparison Project (CMIP) analysis to constrain future projections in ocean carbon inventories requires standardized output of ocean biogeochemical variables (carbon, oxygen, nutrients) across many models and experiments (Wilson et al., 2022). These archived data are often distributed across many federated storage nodes, including duplicated, incomplete or corrupted versions. Although ESGF provides a federated infrastructure for searching and accessing these archives, practical research workflows often require additional automation to determine whether a given combination of variables, experiments, ensemble members, and grid configurations is both scientifically suitable and operationally downloadable before substantial time is spent retrieving files.

This issue is especially important for CMIP-style analyses in which researchers may need to confirm that the same Earth System Model provides paired pre-industrial, historical and ssp-scenario simulations, that multiple variables of interest are available and whether these are separately archived on compatible grids for downstream analyses; and that temporal coverage is continuous across archived files. Manual inspection of catalogue search results can become slow, repetitive, and error-prone when screening many candidate models or variables across multiple ESGF nodes.

Among its features, precog-esgf-intake provides an interactive workflow for shortlisting Earth System Model datasets that satisfy compound criteria across experiments and variables. The software validates temporal coverage, grid consistency, and file availability across published ESGF archives before download. Users can export search results as tabular summaries, inspect

and refine shortlisted datasets, and initiate batch downloads or trigger file integrity checks on locally assigned output paths. Retrieved data are then organised into a directory structure suitable for reproducible downstream analysis. This workflow is particularly useful when researchers need to screen many candidate models and variables before selecting datasets that are both scientifically appropriate and operationally accessible within their computational and storage constraints.

Furthermore, by automating the retrieval and validation of CMIP grid metrics like `volcello`, `areacello` as default (and enabling further bespoke searches for `deptho` or `thkcello` as a validated substitutes when `volcello` is unavailable) — `precog-esgf-intake` reduces the need for users to understand CMIP metadata and ESGF directory conventions. By wrapping ESGF discovery in a simple Python CLI and allowing users to tailor searches by editing a straightforward `search_criteria.toml` configuration file, the package supports both accessible default workflows and more bespoke dataset selection. This lowers the barrier for researchers new to CMIP while enabling experienced users to specify models, experiments, variables, frequencies, and other search constraints. Its lightweight design also supports terminal-based HPC workflows, allowing data discovery and ingestion from visualisation nodes and Jupyter sessions rather than interactive web portals or manual downloads.

State of the field

CMIP data are currently accessible through the ESGF metagrid web application, which provides web-based discovery and download services for CMIP5, CMIP6, CMIP6Plus, and the upcoming CMIP7 phases. Recent development efforts have focused on analysis-ready, cloud-optimised approaches based on in-memory object storage (Mizielinski et al., 2026), with catalogue sweep tools enabling more scalable access through Python and `xarray`, while community evaluation frameworks like `ESMValTool` (Righi et al., 2020) supporting integrated diagnostics for benchmarking model outputs.

`precog-esgf-intake` builds on the ESGF catalogue node sweeping implementation from `intake-esgf` (Collier et al., 2026), but targets a different level of the workflow: rather than focusing only on data access primitives and caching data-responses to in-memory use, it provides an interactive command-line environment for ample discovery, screening, validating and managing downloads of published ESGF archive products that satisfy scientific constraints relevant to Earth system and ocean biogeochemistry applications. The software is intended for situations in which researchers need to move efficiently from broad discovery and catalogue searches to a smaller, analysis-ready subset of model output that satisfies practical and scientific constraints.

Key functionality

The software provides several features tailored to archive-scale Earth system data workflows:

- Definition of ESGF search criteria by modifying a `search_criteria.toml` configuration file.
- Automated search through the ESGF catalogue using project, variable, experiment, and temporal filters (inherited from `intake-esgf`).
- Ensemble-aware screening mode for selecting one validated `piControl` anchor branch and retaining all compatible `Historical` ensemble members on the same grid.
- Single-branch screening mode for selecting one internally consistent `piControl`/`Historical` branch pairing per accepted model/grid.
- Combined-variable screening for retaining models that provide all requested variables simultaneously across both `piControl` and `historical` runs, producing a single shortlist for downstream download.

- 88 ▪ Verification and logging of continuity of date stamps and availability of piControl and
- 89 Historical runs on consistent grids (e.g., regular grid gr and native grid gn).
- 90 ▪ Export of simple Dataframes for realised ESGF catalogue searches.
- 91 ▪ Verification of server responses and flagging shortlisted ESM outputs as 'Downloadable'.
- 92 ▪ Parallelized URL checks for fastest connection in case same data are available across
- 93 different nodes.
- 94 ▪ Parallel batch downloading of files from multiple ESGF nodes with retry and integrity
- 95 checks.
- 96 ▪ Parallel batch search and download of grid cell measures (e.g., areacello and volcello)
- 97 with archive snapshot of relaxed regex matches for later inspection.
- 98 ▪ Local directory layout optimised for downstream analysis.
- 99 ▪ Example Jupyter notebook workflow covering TOML setup, catalogue screening, and
- 100 download preparation.

101 Software design and workflow overview

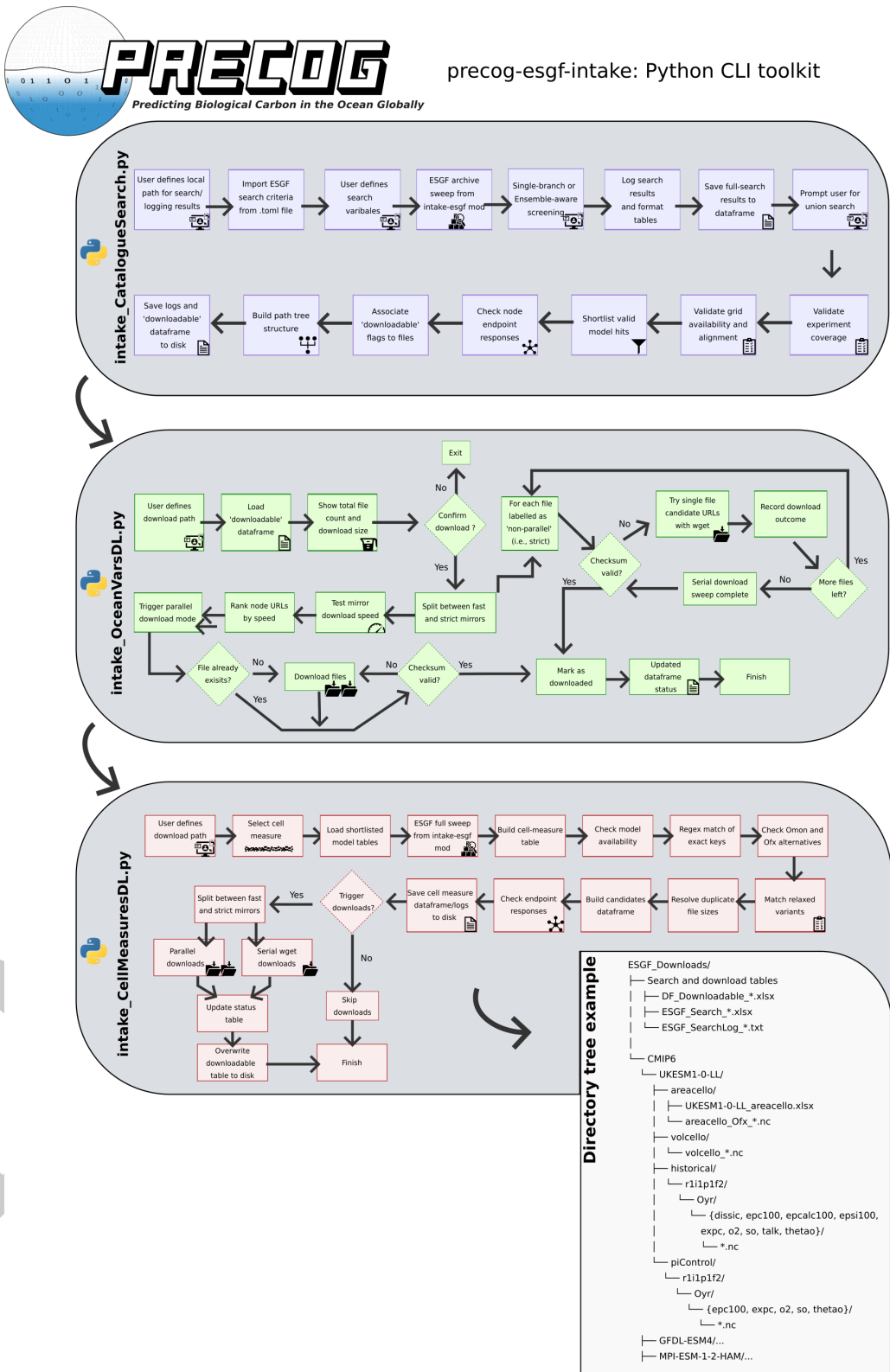
102 precog-esgf-intake implements a staged workflow for archive-scale ESGF data discovery,
 103 screening, and retrieval. Rather than moving directly from catalogue search to cached download,
 104 the software separates archive interrogation, shortlist generation, branch and grid validation,
 105 downloadability checks, and file retrieval into distinct command-line steps, allowing users to
 106 inspect and validate intermediate results before proceeding.

107 In a typical workflow (Figure 1), the user first edits a user-facing `search_criteria.toml`
 108 configuration file to define the intended ESGF search criteria, including project, activity,
 109 experiment, frequency, variables (optional), and grid labels. This externalised configuration
 110 makes search intent easier to reproduce than editing hard-coded dictionaries in source code,
 111 while still allowing interactive entry of `variable_id` when that field is intentionally left blank.
 112 The catalogue search stage then queries ESGF holdings for matching CMIP products (CMIP6 by
 113 default), exports tabular search summaries for inspection, and applies screening logic to retain
 114 scientifically relevant combinations such as paired piControl and historical simulations on
 115 common valid grids.

116 A key feature of the workflow is the ability to **combine** multiple requested variables into a single
 117 conditional shortlist. Rather than screening each variable independently, users can require
 118 that the same model provides all requested variables simultaneously, for both piControl
 119 and historical, before that model progresses to the downloadable shortlist. This combined
 120 screening is particularly useful for multi-field diagnostics where consistent co-availability of
 121 variables is required for downstream interpretation.

122 The workflow supports both single-branch and ensemble-aware catalogue screening. In single-
 123 branch mode, the search retains one internally consistent piControl/historical branch
 124 pairing for each accepted model-grid combination. In ensemble-aware mode, the workflow first
 125 identifies a validated piControl anchor branch and then retains all compatible historical
 126 ensemble members on the same model/grid pair, provided that the requested variables are
 127 complete and continuity checks are satisfied. Grid-level diagnostics are generated from catalogue
 128 metadata so that invalid model-grid combinations can be rejected before download preparation,
 129 and validation outputs are preserved for later auditing.

130 Subsequent stages verify whether shortlisted files are reachable on remote ESGF nodes, assign
 131 local destination paths, and download both target variables and required supporting grid-cell
 132 measures such as areacello and volcello. By treating search results, validation outputs, and
 133 downloadable file lists as explicit intermediate artefacts, the software supports both interactive
 134 exploratory use and more reproducible archive-based Earth system workflows in which dataset
 135 selection decisions can be inspected, traced, and repeated at a later time.



136

137 {width=90%} Figure 1. *precog-esgf-intake* toolkit overview and directory structure of an
 138 example ESGF download. The top-level directory contains search outputs and model-specific
 139 CMIP6 data organised by model, experiment, variable, and annual files.

Research impact and applications

A representative use case is the identification of CMIP6 models that simultaneously provide both pre-industrial and historical outputs for ocean biogeochemical variables such as `expc` and `epc100`, together with auxiliary or supporting variables and the associated grid-cell measures required for downstream analyses. The repository accompanying this software includes a [workflow example](#) demonstrating this type of archive screening and retrieval process.

This is particularly relevant for ocean biogeochemistry and carbon-cycle studies, where analyses often depend on coherent combinations of physical and biogeochemical fields rather than isolated variables (e.g., retrieval of carbonate system fields as well as ocean state physical variables, (Wilson et al., 2022)). In such cases, the time spent screening archive holdings, checking consistency, and organising downloads can be substantial, and purpose-built automation improves both efficiency and reproducibility. This data discovery and retrieval functionality by `precog-esgf-intake` is also complementary to established CMIP preprocessing packages such as `xMIP` (Busecke et al., 2023) and `ESMValTool` (Righi et al., 2020), which provide tools for harmonising, cataloguing, and preparing model outputs for analysis; together, these packages support a reproducible workflow from archive discovery and dataset selection through preprocessing and scientific analysis.

Therefore, `precog-esgf-intake` fills a workflow gap between catalogue access and scientific analysis. The software capabilities support workflows in which data access is itself a significant part of the scientific process, particularly when analyses depend on assembling coherent, scientifically consistent, and analysis-ready ensemble subsets of Earth system model output from distributed archives.

AI usage disclosure

AI assistance was used solely to improve readability of selected software documentation (Claude Sonnet 5 by Anthropic). No AI tools were used to design, implement, or validate the software. All AI-assisted documentation was reviewed and edited by the authors, who retain full responsibility for its accuracy and content. Authors made all the core design and architectural decisions.

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